

232.2.4971 - Quantum computation

Homework Assignment 01

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Due: 09/05/2026 - extended

Question 01 - Inner Products, Angles and Measurement of a single qubit

Let $|\psi\rangle = \cos(\pi/8)|0\rangle + \sin(\pi/8)|1\rangle$.

- Write the state $|\psi\rangle$ using the standard notation. Write $\langle\psi|$ using the standard notation.
- Verify that $\| |\psi\rangle \|^2 \equiv \langle\psi|\psi\rangle = 1$.
- What is the inner product between $|\psi\rangle$ and $|0\rangle$? What is the angle between $|\psi\rangle$ and $|0\rangle$? (Note: the angle between two vectors u, v is defined as $\arccos\left(\frac{\langle u, v \rangle}{\|u\| \cdot \|v\|}\right)$)
- You are given a quantum state which is either $|\psi\rangle$ or $|0\rangle$ uniformly at random. You win if you are able to identify correctly the state that you are given, and you lose otherwise. Calculate the winning probabilities of the following strategies:
 - Measure the qubit in the computational basis. Answer '0' if the outcome is '0', and answer '1' if the outcome is '1'.
 - Apply the Hadamard gate $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, and then measure the qubit in the computational basis. Answer '0' if the outcome is '0', and answer '1' if the outcome is '1'.

0.1 Answer 01

- in standard notation $|\psi\rangle$ is $\begin{bmatrix} \cos(\pi/8) \\ \sin(\pi/8) \end{bmatrix}$, and $\langle\psi|$ is $\begin{bmatrix} \cos(\pi/8) & \sin(\pi/8) \end{bmatrix}$.
- Let's verify $\| |\psi\rangle \|^2 = 1$:

$$\| |\psi\rangle \|^2 = \langle\psi|\psi\rangle = \begin{bmatrix} \cos(\pi/8) & \sin(\pi/8) \end{bmatrix} \begin{bmatrix} \cos(\pi/8) \\ \sin(\pi/8) \end{bmatrix} = \cos^2(\pi/8) + \sin^2(\pi/8) = 1$$

- The inner product between $|\psi\rangle$ and $|0\rangle$ is:

$$\begin{aligned} \langle\psi|0\rangle &= \cos(\pi/8) \langle 0|0\rangle + \sin(\pi/8) \langle 0|1\rangle \\ &= \cos(\pi/8) \end{aligned}$$

The angle is:

$$\begin{aligned} \frac{\langle\psi|0\rangle}{\| |\psi\rangle \| \| |0\rangle \|} &= \frac{\cos(\pi/8)}{|\langle\psi|\psi\rangle|^2 \cdot |\langle 0|0\rangle|^2} \\ &= \frac{\cos(\pi/8)}{|1|^2 \cdot |1|^2} \\ &= \cos(\pi/8) \end{aligned}$$

- (a) Let Π_0 and Π_1 be the projection operator for the computational basis, then the probability of winning is:

$$\begin{aligned} \Pr\{win\} &= \Pr\{win \mid |0\rangle\} \Pr\{|0\rangle\} + \Pr\{win \mid |\psi\rangle\} \Pr\{|\psi\rangle\} \\ &= \langle 0|\Pi_0|0\rangle / 2 + \langle\psi|\Pi_1|\psi\rangle / 2 \\ &= |\langle 0|0\rangle|^2 / 2 + |\langle 1|\psi\rangle|^2 / 2 \\ &= (1 + \sin^2(\pi/8)) / 2 \end{aligned}$$

- (b) Let Π_0 and Π_1 be the projection operator for the computational basis, then the probability of winning is:

$$\begin{aligned}
\Pr\{win\} &= \Pr\{win \mid |0\rangle\} \Pr\{|0\rangle\} + \Pr\{win \mid |\psi\rangle\} \Pr\{|\psi\rangle\} \\
&= \langle 0|H\Pi_1H|0\rangle/2 + \langle \psi|H\Pi_0H|\psi\rangle/2 \\
&= (\langle 0| + \langle 1|) |1\rangle\langle 1| (|0\rangle + |1\rangle)/4 \\
&\quad + ((\cos(\pi/8) + \sin(\pi/8)) \langle 0| + (\cos(\pi/8) - \sin(\pi/8)) \langle 1|) \\
&\quad |0\rangle\langle 0| \\
&\quad (\cos(\pi/8) + \sin(\pi/8)) |0\rangle + (\cos(\pi/8) - \sin(\pi/8)) |1\rangle)/4 \\
&= |\langle 1|1\rangle|^2/4 + |(\cos(\pi/8) + \sin(\pi/8)) \langle 0|0\rangle|^2/4 \\
&= (1 + 1 + 2\cos(\pi/8)\sin(\pi/8))/4 \\
&= 1/2 + \sin(\pi/4)/4
\end{aligned}$$